

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

Third Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

**Course Code: MAT201****Course Name: Partial Differential equations and Complex analysis**

Max. Marks: 100

Duration: 3 Hours

**PART A***Answer all questions. Each question carries 3 marks*

Marks

- 1 Form a partial differential equation from the relation  $z = (x + y)f(x^2 - y^2)$ . (3)
- 2 Solve  $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(3x + 4y)$ . (3)
- 3 Write any three assumptions involved in the derivation of one dimensional wave equation. (3)
- 4 Find the steady state temperature distribution in a rod of length 25 cm, if the ends of the rod are kept at  $20^\circ\text{C}$  and  $70^\circ\text{C}$ . (3)
- 5 Determine whether  $w = \cos z$  is analytic. (3)
- 6 Check whether the function  $xy^2$  is the real part of an analytic function. (3)
- 7 Using Cauchy's integral formula, Evaluate  $\int_C \frac{z^2+1}{z^2-1} dz$  where  $C$  is the circle of unit radius with centre at  $z = 1$ . (3)
- 8 Find the Taylor's series of  $\frac{1}{z}$  about the point  $z = 1$ . (3)
- 9 Find the Laurent series of  $z^2 e^{1/z}$  about  $z = 0$  and determine the region of convergence. (3)
- 10 Find the zeros and their order of the function  $\sin^2(z)$ . (3)

**PART B***Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Find the differential equation of all planes which are at a constant distance 'c' from the origin. (7)
- (b) Solve  $y^2 p - xyq = x(z - 2y)$  (7)

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- 12 (a) Solve  $pq + 2x(y+1)p + y(y+2)q - 2(y+1)z = 0$ . (7)
- (b) Solve  $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$  where  $u(x, 0) = 3e^{5x}$  by method of separation of variables. (7)

**Module 2**

- 13 (a) A tightly stretched string of length one cm is fastened at both ends. Find the displacement of a string if it is released from rest from the position  $\sin \pi x + 5 \sin 3\pi x$ . (7)
- (b) A rod of 30 cm long has its ends A and B kept at  $30^\circ\text{C}$  and  $90^\circ\text{C}$  respectively until steady state temperature prevails. The temperature at each end is then suddenly reduced to zero temperature and kept so. Find the resulting temperature function  $u(x, y)$ , taking  $x = 0$  at A. (7)
- 14 (a) A tightly stretched homogeneous string of unit length with its fixed ends at  $x = 0$  and  $x = 1$  executes transverse vibrations. The initial velocity is zero and the initial deflection is given by  $u(x, 0) = \begin{cases} 1, & 0 \leq x < \frac{1}{2} \\ -1, & \frac{1}{2} \leq x \leq 1 \end{cases}$ . Find the deflection  $u(x, t)$  at any time  $t$ . (7)
- (b) Derive one dimensional heat equation (7)

**Module 3**

- 15 (a) Find the image of the semi circle  $y = \sqrt{4 - x^2}$  under the transformation  $w = z^2$ . (7)
- (b) Show that  $u = x^2 - y^2 - y$  is harmonic. Also find the corresponding harmonic conjugate function. (7)
- 16 (a) Find the image of the circle  $|z - 1| = 1$  under the mapping  $w = \frac{1}{z}$ . (7)
- (b) If  $f(z) = u(x, y) + iv(x, y)$  is analytic and  $uv = 2023$ , then show that  $f(z)$  is a constant (7)

**Module 4**

- 17 (a) Using Cauchy's integral formula, Evaluate the integral  $\int_C \frac{2z+3}{z^2} dz$ , where  $C$  is a circle  $|z - i| = 2$  counter clockwise. (7)
- (b) Evaluate  $\int_C (z^2 + 3z) dz$  along the circle  $|z| = 2$  from  $(2, 0)$  to  $(0, 2)$  in counter-clockwise direction. (7)

- 18 (a) Using Cauchy's integral formula, Evaluate  $\oint_C \frac{e^{2z}}{(z+1)^4} dz$  where C is the circle  $|z| = 2$ . (7)
- (b) Expand  $f(z) = \frac{z+1}{z-1}$  as a Taylor series about  $z = -1$  (7)

**Module 5**

- 19 (a) Find the Laurent series expansion of  $f(z) = \frac{1}{1-z^2}$  about  $z = 1$  in the regions (7)
- (i)  $0 < |z - 1| < 2$  (ii)  $|z - 1| > 2$
- (b) Evaluate  $\int_0^{2\pi} \frac{1}{2+\cos\theta} d\theta$ . (7)
- 20 (a) Using Cauchy's Residue theorem, Evaluate  $\int_C \frac{30z^2-23z+5}{(2z-1)^2(3z-1)} dz$  where C is the circle  $|z| = 1$  counter-clockwise. (7)
- (b) Using contour integration, evaluate  $\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^3} dx$ . (7)

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